

Causal Foundation Models (Title will change)

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1 Introduction

2 Background and related work

2.1 Structural causal model and interventions

We work with an observational dataset $\mathcal{D}_{\text{obs}} = \{(X_i, T_i, Y_i)\}_{i=1}^n$, where each row records a unit's covariates $X_i \in \mathbb{R}^d$, the treatment it actually received $T_i \in \{0, 1\}$, and the outcome observed under that treatment, $Y_i \in \mathbb{R}$.

We assume $\mathcal{D}_{\text{obs}}$ is generated by a structural causal model (SCM) $\psi = (G, f, p_\varepsilon)$ over $\{X, T, Y\}$: G a directed graph, $f = \{f_V\}$ structural equations shared across all units, p_ε the exogenous noise distribution. Each row of $\mathcal{D}_{\text{obs}}$ is one sample $\varepsilon_i \sim p_\varepsilon$ propagated through f .

An intervention $\text{do}(T = t)$ under SCMs sets the treatment to value t and severs its dependence on its normal causal parents, while leaving every other structural equation untouched. This defines the potential outcome:

$$Y^{\text{do}(t)}(x, \varepsilon) = f_Y(x, t, \text{pa}_{-T}(\psi; x, \varepsilon), \varepsilon_Y), \quad t \in \{0, 1\}.$$

where $\text{pa}_{-T}(\psi; x, \varepsilon)$ denotes the values of Y 's graph-parents other than T , propagated through ψ .

We assume $Y^{\text{do}(0)}$ and $Y^{\text{do}(1)}$ share the same noise draw ε , so they are *coupled* rather than independent — this coupling is what a paired observation carries and a single-outcome observation does not. Because the generating mechanism is fully specified for any SCM used as a synthetic data source, this coupling lets us sample both potential outcomes of a synthetic unit together:

Assumption 1 (Paired-outcome sampling). *For every x in the support of X , $\mathbb{E}_{p_\varepsilon}[Y^{\text{do}(t)}(x, \varepsilon)^2] < \infty$ for $t \in \{0, 1\}$, and the SCM admits paired-outcome sampling: a single draw $\varepsilon \sim p_\varepsilon$ yields both $Y^{\text{do}(0)}(x, \varepsilon)$ and $Y^{\text{do}(1)}(x, \varepsilon)$.*

The conditional average treatment effect (CATE) at covariate value x is

$$\tau(x) := \mathbb{E}[Y^{\text{do}(1)} - Y^{\text{do}(0)} \mid X = x],$$

and the average treatment effect (ATE) is its average over the covariate population, $\mathbb{E}_X[\tau(X)]$.

general version of MALC can be introduced here, or beginning of section 3. We specify how MALC is used/adjusted inline when we need it No need for the MALC here, we can discuss why that is in a meeting once we complete the first draft.

2.2 Prior-data fitted networks for Bayesian inference

We define Bayesian prediction as the task of computing the posterior predictive distribution (PPD), $p(y \mid x, \mathcal{D})$, for a query x given an observed dataset $\mathcal{D}$:

$$p(y \mid x, \mathcal{D}) = \int_{\Theta} p(y \mid x, \theta) p(\theta \mid \mathcal{D}) d\theta \quad (1)$$

where predictions are averaged over a hypothesis space Θ , weighted by the posterior belief $p(\theta \mid \mathcal{D}) \propto p(\mathcal{D} \mid \theta)p(\theta)$.

Instead of evaluating this integral anew for every dataset — intractable outside conjugate special cases — Prior-Data Fitted Networks (PFNs) amortize the computation: a neural network q_{ϕ} is trained once, offline, on synthetic $(\theta, \mathcal{D})$ pairs sampled from the prior $p(\theta)$, directly approximating the PPD. At inference time, a real dataset $\mathcal{D}_{\text{real}}$ is provided as context, and the network outputs an approximation to $p(y \mid x, \mathcal{D}_{\text{real}})$ in a single forward pass without gradient updates.

2.3 Causal foundation models

Causal foundation models (CFMs) apply the PFN paradigm to causal effect estimation. In this setting, the hypothesis space consists of structural causal models (SCMs) $\psi \sim p(\psi)$. Given covariates x and an observational dataset $\mathcal{D}_{\text{obs}}$, a CFM parameterised by ϕ produces an amortized posterior predictive distribution

$$q_{\phi}(Y \mid \mathcal{D}_{\text{obs}}, x, \text{do}(t)) \quad (2)$$

for each intervention value $t \in \{0, 1\}$.

DoPFN, UWYK, CausalPFN, CausalFM

3 Methodology

Current causal foundation models [?] train a single-outcome head on (T_i, Y_i) pairs, using only the outcome under the treatment assigned to each unit. This convention discards half of the information relevant to estimation and limits the effective capacity of the model. We replace the single-outcome head with one that outputs the joint density over the paired potential outcomes; two consequences follow. Section 3.1 establishes that the joint objective halves the asymptotic mean-squared error of every marginal density and, by linearity, of the CATE and ATE estimators themselves. Section 3.2 shows that the joint density recovers dependence structure between $Y_{\text{do}(0)}$ and $Y_{\text{do}(1)}$ that no pair of marginals can express, enabling distributional inference over CATE and ATE that a marginal head can produce only conservatively.

3.1 Statistical efficiency of the joint objective

For each query covariate X with observational context $\mathcal{D}_{\text{obs}} = \{(X_i, T_i, Y_i)\}_{i=1}^N$, the joint head predicts a density $f(y_0, y_1 \mid X, \mathcal{D}_{\text{obs}})$ on $\mathbb{R}^2$. The training loss is the negative log-density against paired labels drawn from the SCM prior,

$$\mathcal{L}(y_0^*, y_1^* \mid X, \mathcal{D}_{\text{obs}}) = -\log f(y_0^*, y_1^* \mid X, \mathcal{D}_{\text{obs}}), \quad (3)$$

averaged over queries in a batch. Positivity of f on all of $\mathbb{R}^2$ is required for the loss to remain finite. We enforce it by combining a bounded interior $K \times K$ grid on $\mathcal{I} = [-1, 1]^2$ with Gaussian tails

along the four extended-axis and four diagonal directions, producing a nine-region mixture. This construction requires $K^2 + 9 + 4$ trainable parameters per query, and at the values of K used by marginal-only head models the quadratic dependence would explode the computational cost. We therefore train at $K = \sqrt{J}$, where J is the baseline marginal-only model’s grid. Naively reducing the per-dimension resolution, however, shrinks the model’s representational capacity, uniformly degrading its estimation quality across both point and distributional metrics.

We overcome this issue by applying MALC [1] at inference. MALC is a mixture-of-log-concave estimator over uniformly binned data that produces both point and density estimations. It recovers the true values on wider grids—smaller K —up to a small bounded error, provided the per-bin probabilities are themselves estimated with sufficient accuracy. Because the head is trained under that objective, $K = \sqrt{J}$ suffices to preserve the model’s representational capacity, and the joint head’s parameter count then matches that of a marginal-only, preserving the computational cost. Appendices A and C detail the construction of the joint head and the MALC procedure (base class, mixture order-selection criterion, per-query inference cost), respectively.

With computational cost matched and representational capacity addressed, we examine the statistical efficiency of the joint objective. Lemma 1 establishes that the joint head is strictly more efficient than the marginal one: the objective in (3) retains at least all the information available to the marginal objective.

Lemma 1. *Let $Z_i^R := (X_i, Y_{\text{do}(0)}(x_i, \varepsilon_i), Y_{\text{do}(1)}(x_i, \varepsilon_i))$ denote the per-unit observation under the joint objective and $Z_i^U := (X_i, T_i, Y_{T_i}(x_i, \varepsilon_i))$ the per-unit observation under the marginal objective, with T_i drawn from any distribution on $\{0, 1\}$ independent of ε_i . Then Z^U is a randomised function of Z^R : sampling T from the same distribution and setting $Z^U = (X, T, Y_T)$ recovers the law of the single-outcome observation from that of the paired one. Consequently any parameter of the joint law $\mathbb{P}^*(Y_{\text{do}(0)}, Y_{\text{do}(1)} | X)$ estimable from Z^U is estimable from Z^R at least as well.*

Because the treatment, T_i , is randomly assigned, the marginal observation contains information about $Y_{\text{do}(t)}$ only for the t selected by T_i ; the joint observation, by contrast, contains information about both potential outcomes at every unit. The Fisher information the marginal objective discards therefore depends on the distribution of T_i . Theorem 2 shows that this loss propagates, via the Cramér–Rao bound, to the mean-squared error of every marginal density and, by linearity, to the CATE estimator itself.

Theorem 2. *Let $T_i \sim \text{Bernoulli}(\pi)$ with $\pi \in (0, 1)$, write $\pi_1 = \pi$ and $\pi_0 = 1 - \pi$, and assume standard MLE regularity. For every query x :*

(i) *for each $t \in \{0, 1\}$, the marginal density estimators satisfy*

$$\lim_{N \rightarrow \infty} N \cdot \mathbb{E}[(\hat{p}_U(y | x, t) - p^*(y | x, t))^2] = \frac{1}{\pi_t} \cdot \lim_{N \rightarrow \infty} N \cdot \mathbb{E}[(\hat{p}_R(y | x, t) - p^*(y | x, t))^2]; \quad (4)$$

(ii) *the CATE estimator $\hat{\tau}(x) = \hat{\mu}_1(x) - \hat{\mu}_0(x)$ satisfies*

$$\lim_{N \rightarrow \infty} \frac{\mathbb{E}[(\hat{\tau}^U(x) - \tau(x))^2]}{\mathbb{E}[(\hat{\tau}^R(x) - \tau(x))^2]} = \frac{1}{2\pi(1 - \pi)}. \quad (5)$$

The ratio in (5) rescales the whole limiting error distribution. The Corollary 3 records this second consequence: the joint objective is more stable across problems, not merely more accurate on average.

Corollary 3. *Under the same conditions,*

$$\lim_{N \rightarrow \infty} \frac{\text{Std}_{\text{SCMs}}(\sqrt{\text{PEHE}}^{\text{R}})}{\text{Std}_{\text{SCMs}}(\sqrt{\text{PEHE}}^{\text{U}})} = \sqrt{2\pi(1 - \pi)}.$$

Proofs of Lemma 1, Theorem 2 and Corollary 3 are given in Appendix B.

The results are asymptotic in N and do not depend on the covariate dimension d , though larger d requires larger N before the limit becomes informative. Three conditions can nevertheless prevent the theory from materialising in practice. First, the Cramér–Rao bound governs variance under maximum likelihood, so training under a non-MLE proper-scoring loss may preserve the qualitative ordering while losing the predicted magnitude of the improvement. Second, the ordering relies on the estimator converging to the true marginal density; if the joint parameterisation approximates the marginal strictly worse than the marginal-only one, the ordering can flip. Third, the predicted improvement is a reduction in variance, and the theory assumes the asymptotic regime in which bias does not dominate the error; when it does, the improvement may not be visible in practice. We therefore verify the theory empirically in Section 4, reporting controlled simulations that measure whether the improvement materialises, the rate of asymptotic convergence, and how the sample complexity scales with d .

3.2 Distributional inference for CATE and ATE

The primary purpose of any model, including tabular foundation models, is to provide information to the user for a better-informed decision. For users interested merely in the observational distribution, models such as TabPFN ? deliver sufficient information about the variable of interest by reporting both its density and a scalar point estimate. Existing CFMs, however, fall short of this standard for users interested in the interventional distribution, and report only a limited view: some provide a scalar CATE estimate together with a distribution over responses under each treatment ?, while others provide a scalar CATE with a distribution over the treatment effect but no response information ?. To our knowledge, no existing method delivers the response and CATE distributions simultaneously, and no existing method delivers an ATE distribution at all. A decision maker using a CFM, however, generally requires all three: the response densities support outcome decisions (e.g., whether a treated unit’s outcome exceeds a clinical or economic threshold), while the CATE and ATE densities support the treatment decision itself (e.g., whether to intervene at all). One might argue that a point-wise CATE estimate combined with the marginal densities already provides sufficient information for this decision; however, same point estimate and confidence interval can arise from different shapes of uncertainty that call for different actions. For example, a near-uniform effect density and a bimodal density with two contrasting (negative and positive) modes can yield identical point estimates and confidence intervals: the uniform density indicates that no action can be preferred over another, whereas the bimodal density indicates that with domain knowledge one may favour one mode over the other.

To recover the missing distributional information, one might attempt to reconstruct a CATE distribution from the marginals, but this is possible only under an independence assumption between $Y_{\text{do}(0)}$ and $Y_{\text{do}(1)}$, which is generally false, and discarding the coupling yields spurious treatment-effect values. The joint head resolves this gap and simultaneously refines the marginal response distribution that current models already provide; it therefore both improves what existing models report and delivers the additional information decision makers require.

Sections 3.2.1 and 3.2.2 describe how the per-unit CATE density and the population-level ATE density are recovered from the joint head’s output.

3.2.1 Per-unit CATE distribution via MALC

The joint head produces a discrete distribution on a $K \times K$ grid with mass p_{ij} at bin centre (y_0^i, y_1^j) . MALC fits a continuous joint density $\hat{p}(y_0, y_1)$ to these bin masses as a mixture of log-concave components. The per-unit CATE density then follows by the change of variables $(Y_0, Y_1) \mapsto (Y_0, \tau)$ with $\tau := Y_{\text{do}(1)} - Y_{\text{do}(0)}$, marginalising out Y_0 :

$$f_\tau(t) = \int \hat{p}(y_0, y_0 + t) dy_0. \quad (6)$$

Danisman [2026] establishes that any 1D mean derived from a MALC fit differs from the true mean by at most the grid width δ in the limit $N \rightarrow \infty$. Since CATE is the difference of the two marginal means, the triangle inequality yields the corresponding CATE-mean bound

$$|\hat{\mu}_\tau - \mu_\tau| \leq 2\delta \quad \text{as } N \rightarrow \infty,$$

where $\hat{\mu}_\tau := \int t f_\tau(t) dt$ and μ_τ denotes the true CATE mean. Since $\delta = 2/K$, at the reduced resolution $K = \sqrt{J}$ of Section 3.1 the CATE-mean error is asymptotically bounded by $4/\sqrt{J}$. The cost of the reduced resolution therefore vanishes as the context size and baseline resolution grow, leaving the joint head’s efficiency gain as the net effect. The full MALC fitting procedure is given in Appendix C.

3.2.2 ATE distribution via Optimal Transport

Given per-query CATE densities $\{f_\tau^{(n)}\}_{n=1}^N$ across a covariate sample $\{X^{(n)}\}_{n=1}^N$, the ATE density f_{ATE} is defined as their 2-Wasserstein barycenter with uniform weights. For univariate densities on a common support this barycenter admits the closed-form quantile average

$$Q_{\text{ATE}}(u) = \frac{1}{N} \sum_{n=1}^N Q_\tau^{(n)}(u), \quad u \in (0, 1), \quad (7)$$

where $Q_\tau^{(n)} = (F_\tau^{(n)})^{-1}$ is the quantile function of the n -th per-query CATE density obtained via MALC. Derivation and the grid-level algorithm are given in Appendix D.

4 Experiments

To isolate the effect of our changes from any confounding factor, we hold the training data and training algorithm of the base CFMs fixed and modify only the head and its inference procedure. We let models achieve their full capacity without increasing training time or per-query inference cost relative to their single-outcome baselines. Although several CFMs are publicly available for inference, retraining is required to change the head, and this limits the set of models we can modify to those whose training infrastructure is public and whose compute budget is tractable.

[Change this part completely] We therefore restrict our study to UWYK [?] and Do-PFN [?], both of which publish their full training pipeline. We train both models with our joint head from scratch on a single H100 with 53 and 30 hours for UWYK and Do-PFN respectively. In both cases the joint

head changes the parameter count by less than 3.5% of the total something something. Step size and training time is matched for UWYK, however, for Do-PFN step size is not publicly available, hence it is possible that we overtrain it. Hence, we provide the values of 10 hours training in Appendix. [\[Up to here.\]](#)

[\[Subject to change\]](#) The two other major CFMs, CausalFM [?] and CausalPFN [?], are excluded on practical grounds: the former’s training cost exceeds our compute budget, and the latter’s training infrastructure is not publicly released. Extending our modifications to larger or closed models is left to future work.

[\[Subject to change\]](#) For the remainder of this section, we denote each modified model with the suffix **-2DMALC** and use it as a drop-in replacement for its base model. UWYK has two potential base variants — one that uses graph-adjacency information during training and one that does not — either of which can serve as the base for 2DMALC. Under our compute budget we adopt only the no-graph variant; we are further motivated by the question of whether the head change on this baseline can nonetheless exceed the performance of the graph-trained variant with full graph information. For Do-PFN, since it ships in a single variant, this problem does not arise.

We evaluate our changes on four fronts. Sections 4.2, 4.3, and 4.4 evaluate point-wise, marginal-density, and CATE/ATE density estimation, respectively, on the RealCause benchmark. Section 4.5 tests Theorems 2–3 on a controlled synthetic prior, tracking the models’ asymptotic behaviour as context size, and covariate dimension vary.

4.1 Evaluation metrics

We report three metrics: two for point estimates and one for distributional predictions.

Point estimates. Per-unit accuracy on the CATE is measured by

$$\sqrt{\text{PEHE}} := \left(\mathbb{E}_X [(\hat{\tau}(X) - \tau(X))^2] \right)^{1/2}, \quad (8)$$

and population accuracy on the ATE by the relative error

$$\varepsilon_{\text{ATE}} := \frac{|\widehat{\text{ATE}} - \text{ATE}|}{|\text{ATE}|}. \quad (9)$$

Both expectations are estimated by empirical averages over held-out queries.

Distributional predictions. Predicted densities are compared to the analytic ground truth f^* under the L^2 distance $L_2(\hat{f}, f^*) = \|\hat{f} - f^*\|_2$, evaluated on a common grid to which every method’s output is resampled and renormalised to integrate to one. We apply it to three targets: the two arm marginals $p(y | X, \text{do}(t))$, the per-unit CATE density $p(\tau | X)$, and the population ATE density. Neither the CATE nor the ATE density is a native output of any baseline we compare. For the CATE density, models without a joint head cannot express $p(\tau | X)$ directly; we recover it from the outer product of the two marginals, which is the independence assumption these models implicitly make. For the ATE density, no method produces one directly; we obtain it, uniformly across methods, as the 2-Wasserstein barycenter of the per-query CATE densities (Section 3.2.2, Appendix D).

4.2 RealCause benchmarks - [A better name is needed]

We investigate whether the modifications improve the baseline models on the semi-synthetic RealCause benchmark [?]. Table 1 reports per-unit error ($\sqrt{\text{PEHE}}$) and population error (ε_{ATE}) across the five datasets.

[Use the updated results!] UWYK-2DMALC is trained on the no-graph variant of UWYK, so the direct comparison is against UWYK No-Ancestral. Across the five datasets, the head change reduces per-unit error by 2–31% and population error by 6–72%. It further outperforms the Full Ancestral variant on all ten cells of the benchmark, indicating that the modification alone, with no graph information, recovers more signal than the base model does even with complete graph information.

The same pattern holds for Do-PFN: the modified head reduces population error by 9–77% on all five datasets and per-unit error by 7–22% on three of the five, with small negative differences (2–4%) on the remaining two.

The head change also affects the current CFM ranking, under the caveat that the modification is applied only to these two base models. Neither UWYK No-Ancestral nor Do-PFN previously outperformed CausalFM, whereas the modified variants now do so on population error across all five datasets and on per-unit error on four of the five. Applying the same modification to CausalFM would likely produce comparable gains. We therefore make no claim of a permanent change in the ranking, but it establishes that models adopting the modification will differ substantively from those that do not.

4.3 Marginal density estimation

4.4 Distributional inference for CATE and ATE

4.5 Context-size and asymptoticity - A better name is needed

Robustness across d and ρ .

- RealCause
- Marginal Plot difference
- CATE + ATE Plots
- N simulation
- d simulation - Appendix

5 Discussion

Table 1: RealCause benchmark

Method	IHDP		ACIC		CPS		PSID		PSID (balanced)	
	$\sqrt{\text{PEHE}} \downarrow$	$\varepsilon_{\text{ATE}} \downarrow$	$\sqrt{\text{PEHE}} \downarrow$	$\varepsilon_{\text{ATE}} \downarrow$	$\sqrt{\text{PEHE}} \downarrow$	$\varepsilon_{\text{ATE}} \downarrow$	$\sqrt{\text{PEHE}} \downarrow$	$\varepsilon_{\text{ATE}} \downarrow$	$\sqrt{\text{PEHE}} \downarrow$	$\varepsilon_{\text{ATE}} \downarrow$
UWYK No Ancestral	6.28 $\pm$ 0.79	0.67 $\pm$ 0.48	3.27 $\pm$ 0.45	0.35 $\pm$ 0.17	13057 $\pm$ 23	1.08 $\pm$ 0.04	21896 $\pm$ 137	0.96 $\pm$ 0.01	21886 $\pm$ 137	0.93 $\pm$ 0.04
UWYK-2DMALC	4.32 $\pm$ 0.57	0.44 $\pm$ 0.07	2.82 $\pm$ 0.46	0.15 $\pm$ 0.08	12826 $\pm$ 18	1.02 $\pm$ 0.04	21992 $\pm$ 131	0.91 $\pm$ 0.00	18463 $\pm$ 157	0.49 $\pm$ 0.01
UWYK Full Ancestral	5.49 $\pm$ 0.78	0.49 $\pm$ 0.79	2.73 $\pm$ 0.44	0.16 $\pm$ 0.11	12938 $\pm$ 23	1.07 $\pm$ 0.03	19710 $\pm$ 230	0.92 $\pm$ 0.07	19722 $\pm$ 233	0.66 $\pm$ 0.17
Do-PFN	6.07 $\pm$ 0.90	0.93 $\pm$ 3.98	4.11 $\pm$ 0.55	0.66 $\pm$ 0.12	12015 $\pm$ 32	0.88 $\pm$ 0.06	20907 $\pm$ 138	0.93 $\pm$ 0.07	22773 $\pm$ 174	1.08 $\pm$ 0.13
Do-PFN-2DMALC	5.06 $\pm$ 0.71	0.53 $\pm$ 0.18	3.15 $\pm$ 0.64	0.29 $\pm$ 0.08	10715 $\pm$ 16	0.12 $\pm$ 0.01	18617 $\pm$ 146	0.44 $\pm$ 0.01	18711 $\pm$ 158	0.44 $\pm$ 0.01
CausalFM	5.86 $\pm$ 7.25	0.70 $\pm$ 1.23	3.30 $\pm$ 1.59	0.33 $\pm$ 0.11	12400 $\pm$ 157	0.94 $\pm$ 0.00	22436 $\pm$ 1296	0.95 $\pm$ 0.01	21071 $\pm$ 1339	0.86 $\pm$ 0.05
CausalPFN	0.58 $\pm$ 0.07	0.03 $\pm$ 0.00	0.92 $\pm$ 0.11	0.06 $\pm$ 0.01	8956 $\pm$ 21	0.14 $\pm$ 0.01	14402 $\pm$ 198	0.22 $\pm$ 0.02	14833 $\pm$ 240	0.26 $\pm$ 0.02

Table 2: [Delta change- show.] Context-size sweep on the CausalPFN polynomial prior, held out from our training corpus. 500 SCMs per context size N . UWYK Baseline is the paper’s separately-trained unconditional checkpoint. Metric: $\sqrt{\text{PEHE}}$ (mean $\pm$ std across SCMs). Bold = best in column.

Method	$N = 50$	$N = 250$	$N = 500$	$N = 1000$	$N = 5000$
UWYK No Ancestral	1.48 $\pm$ 0.25	1.42 $\pm$ 0.23	1.38 $\pm$ 0.26	1.35 $\pm$ 0.24	1.27 $\pm$ 0.24
UWYK-2DMALC	1.25 $\pm$ 0.24	1.11 $\pm$ 0.22	1.09 $\pm$ 0.23	1.06 $\pm$ 0.22	0.92 $\pm$ 0.19
UWYK Full Ancestral	1.21 $\pm$ 0.26	0.94 $\pm$ 0.28	0.87 $\pm$ 0.28	0.80 $\pm$ 0.27	0.69 $\pm$ 0.21
Do-PFN	1.36 $\pm$ 0.26	1.30 $\pm$ 0.25	1.29 $\pm$ 0.29	1.27 $\pm$ 0.27	1.29 $\pm$ 0.28
Do-PFN-2DMALC	1.29 $\pm$ 0.29	1.15 $\pm$ 0.30	1.12 $\pm$ 0.33	1.12 $\pm$ 0.29	1.08 $\pm$ 0.31

A Joint head architecture

The joint head defined in Section 3.1 must return a finite density at every point of $\mathbb{R}^2$ so that the log-loss in (3) is well-posed on unbounded outcomes, while remaining expressive enough on the bounded region $\mathcal{I} = [-1, 1]^2$ ¹. We meet both requirements with a single mixture that places a $K \times K$ histogram on $\mathcal{I}$ and Gaussian tails outside it. The construction below provide the nine regions of the mixture, the $K^2 + 9 + 4$ raw outputs the transformer emits per query, the link functions that map raw outputs to a valid density, and the approximation the design introduces.

Grid and indexing. Partition $\mathcal{I}$ into a uniform $K \times K$ grid of square bins of width $\delta := 2/K$. Bins are indexed by $(j_0, j_1) \in \{0, \dots, K-1\}^2$; bin B_{j_0, j_1} is the $\delta \times \delta$ square with lower-left corner $(-1 + j_0\delta, -1 + j_1\delta)$. For any $y \in [-1, 1]$ and coordinate $k \in \{0, 1\}$, the containing bin index along that coordinate is

$$j_k(y) := \lfloor (y + 1)/\delta \rfloor. \quad (10)$$

Per-query outputs. For each query the transformer emits $K^2 + 9 + 4$ real numbers, grouped into three tensors mapped to normalised parameters by fixed link functions:

- *Interior logits* $\ell \in \mathbb{R}^{K^2}$ softmaxed into a $K \times K$ probability table over the bins,

$$p_{j_0, j_1}(X, \mathcal{D}_{\text{obs}}) = \frac{\exp(\ell_{j_0, j_1})}{\sum_{i_0, i_1} \exp(\ell_{i_0, i_1})}; \quad (11)$$

- *Region logits* $u \in \mathbb{R}^9$ softmaxed into non-negative mixture weights across the nine regions defined below,

$$w_r(X, \mathcal{D}_{\text{obs}}) = \frac{\exp(u_r)}{\sum_{s=1}^9 \exp(u_s)}, \quad r \in \{1, \dots, 9\}; \quad (12)$$

- *Raw tail scales* $(\sigma_L^{(0), \text{raw}}, \sigma_R^{(0), \text{raw}}, \sigma_L^{(1), \text{raw}}, \sigma_R^{(1), \text{raw}})$, one for each coordinate–direction pair, passed through softplus with a small floor $\varepsilon > 0$ and scaled by δ ,

$$\sigma_d^{(k)}(X, \mathcal{D}_{\text{obs}}) = \delta \cdot (\text{softplus}(\sigma_d^{(k), \text{raw}}) + \varepsilon), \quad k \in \{0, 1\}, d \in \{L, R\}. \quad (13)$$

The δ prefactor in (13) makes the tail scales compatible with the interior resolution: doubling K halves both the bin width and the initial tail width, so the density is invariant under resolution changes at initialisation.

Nine regions. The plane $\mathbb{R}^2$ is partitioned into nine regions in three groups: (1) the interior square $\mathcal{I}$, where both $y_0, y_1 \in [-1, 1]$; (2) four edge strips, where exactly one of y_0, y_1 lies outside $[-1, 1]$; and (3) four corner quadrants, where both lie outside. Every query point by default falls in one region, so at inference the mixture weight w_r selects a single component; only its density contributes to f and to $\log f$.

Interior region ($y_0, y_1 \in [-1, 1]$). The histogram is converted to a piecewise-constant density by dividing by the bin area,

$$f_{\text{inner}}(y_0, y_1 \mid X, \mathcal{D}_{\text{obs}}) = \frac{p_{j_0(y_0), j_1(y_1)}}{\delta^2}. \quad (14)$$

The δ^2 factor is the Jacobian from bin probability to density on $[-1, 1]^2$.

¹Outcomes are rescaled per task so that the observed range of Y_T maps to $[-1, 1]$.

Edge regions ($y_i \in [-1, 1]$, $y_{1-i} \notin [-1, 1]$). Consider the representative subcase $y_0 \in [-1, 1]$ and $y_1 < -1$; the other three edges are symmetric. We factor the edge density as $p(y_1)p(y_0 | y_1)$, model the marginal $p(y_1)$ by a half-Gaussian anchored at the boundary, and approximate the conditional $p(y_0 | y_1)$ by the inner region at the boundary:

$$f_{\text{edge}}(y_0, y_1 | X, \mathcal{D}_{\text{obs}}) = p_L^{(1)}(y_1) g(y_0 | y_1 = -1), \quad (15)$$

with

$$p_L^{(1)}(y_1) = \frac{2}{\sigma_L^{(1)}} \phi\left(\frac{y_1 + 1}{\sigma_L^{(1)}}\right), \quad g(y_0 | y_1 = -1) = \frac{p_{j_0(y_0), 0}}{(\sum_{i_0} p_{i_0, 0}) \delta}, \quad (16)$$

where ϕ is the standard normal density.

Holding $p(y_0 | y_1)$ fixed at $y_1 = -1$ is exact approximation when $Y_0 \perp Y_1$ and otherwise introduces a bias that grows with the strength of dependence and the distance past the boundary; for a bivariate normal with correlation ρ and tail scale σ the excess NLL per edge observation is approximately $\rho^2 \sigma^2 / \pi(1 - \rho^2)$. Two effects keep this small in practice: (i) the half-Gaussian factor $p_L^{(1)}$ assigns exponentially decreasing weight to y_1 far from the boundary, so the approximation is applied most heavily where it is most accurate; and (ii) the tail scales $\sigma_d^{(k)}$ are trained end-to-end and shrink whenever the interior distribution concentrates away from the boundary, further suppressing the region where the approximation would matter.

Corner regions ($y_0, y_1 \notin [-1, 1]$). Let $(s_0, s_1) \in \{L, R\}^2$ select the quadrant with corner $(c_0, c_1) \in \{-1, +1\}^2$. The corner density is a bivariate normal centred at the corner, with correlation ρ (defined below) and standardised coordinates $\tilde{y}_k = (y_k - c_k) / \sigma_{s_k}^{(k)}$:

$$f_{\text{corner}}(y_0, y_1 | X, \mathcal{D}_{\text{obs}}) = \frac{1}{Z(s_0, s_1, \rho)} \phi_2(\tilde{y}_0, \tilde{y}_1; \rho). \quad (17)$$

The truncation constant Z is the probability mass the bivariate normal places in the quadrant selected by (s_0, s_1) ; by Sheppard's identity,

$$Z(s_0, s_1, \rho) = \begin{cases} \frac{1}{4} + \frac{\arcsin \rho}{2\pi} & \text{if } s_0 = s_1, \\ \frac{1}{4} - \frac{\arcsin \rho}{2\pi} & \text{if } s_0 \neq s_1. \end{cases} \quad (18)$$

Correlation inherited from the interior. The correlation ρ in (17)–(18) is not a separate trained parameter by design choice. It is defined as the Pearson correlation of the discrete distribution $\{p_{j_0, j_1}\}$ evaluated at bin midpoints $m_k(j_k) := -1 + \delta(j_k + \frac{1}{2})$:

$$\begin{aligned} \bar{y}_k &= \sum_{j_0, j_1} p_{j_0, j_1} m_k(j_k), \\ v_k &= \sum_{j_0, j_1} p_{j_0, j_1} (m_k(j_k) - \bar{y}_k)^2, \\ c &= \sum_{j_0, j_1} p_{j_0, j_1} (m_0(j_0) - \bar{y}_0)(m_1(j_1) - \bar{y}_1), \\ \rho &= \frac{c}{\sqrt{v_0 v_1 + \varepsilon}}. \end{aligned} \quad (19)$$

Tying ρ to the interior grid removes one degree of freedom, keeps the corner coupling consistent with what the interior region already represents, and prevents the tail and interior components from disagreeing about the dependence between $Y_{\text{do}(0)}$ and $Y_{\text{do}(1)}$.

Full mixture and training loss. The joint density is

$$f(y_0, y_1 \mid X, \mathcal{D}_{\text{obs}}) = w_1 f_{\text{inner}} + \sum_{r=2}^5 w_r f_{\text{edge}}^{(r)} + \sum_{r=6}^9 w_r f_{\text{corner}}^{(r)}, \quad (20)$$

and the loss in (3) is evaluated at the region of the observed label (y_0^*, y_1^*) ; the region indicator selects a single component, so only its weight and parameters receive gradient at that point. Because the region assignment is deterministic in the label, the mixture weights w are trained purely from which region observations fall into, while the interior grid, tail scales, and derived correlation are trained from where within each region.

B Proofs

In this Section we report the proofs of Lemma 1, Theorem 2, and Corollary 3. To avoid collision with the interior square $\mathcal{I} = [-1, 1]^2$ notation of Appendix A, we denote the Fisher information matrix evaluated at the true joint law $\mathbb{P}^*$ as $\mathbf{J}$ in this Section.

B.1 Proof of Lemma 1

Proof. Let λ denote the law of T on $\{0, 1\}$, chosen independently of ε_i . Define the measurable map $\Phi(x, y_0, y_1; t) := (x, t, y_t)$; then for any Z^{R} with joint law $\mathbb{P}^*$,

$$Z^{\text{U}} \stackrel{d}{=} \Phi(Z^{\text{R}}; T), \quad T \sim \lambda \perp Z^{\text{R}},$$

recovers the law of the marginal observation from the joint one. The map is measurable and marginalises over T without reference to $\mathbb{P}^*$, so Z^{R} is a sufficient statistic for every parameter of $\mathbb{P}^*$. By the Rao–Blackwell theorem, any $\mathbb{P}^*$ -estimator based on Z^{U} is dominated in mean-squared error by its conditional expectation given Z^{R} ; equivalently, any parameter estimable from Z^{U} is estimable from Z^{R} at least as well. $\square$

B.2 Proof of Theorem 2

The two parts share a common Fisher-information calculation.

Proof of part (i). Fix a query point x and a treatment level $t \in \{0, 1\}$, and let ϕ_t parameterise the marginal density $p^*(y \mid x, t)$. Under the joint objective, each unit contributes an observation $(X_i, Y_{\text{do}(0)}(x_i, \varepsilon_i), Y_{\text{do}(1)}(x_i, \varepsilon_i))$; the arm- t coordinate is observed at every unit, so the per-unit Fisher information for ϕ_t equals $\mathbf{J}_{Y_{\text{do}(t)} \mid X}(\phi_t)$.

Under the marginal objective, the arm- t coordinate is observed only when $T_i = t$, which occurs with probability π_t . Since T_i is independent of ε_i , the arm- $(1 - t)$ observations contribute zero information about ϕ_t , and the per-unit Fisher information decomposes as

$$\mathbf{J}_{\text{U}}(\phi_t) = \pi_t \mathbf{J}_{Y_{\text{do}(t)} \mid X}(\phi_t) + (1 - \pi_t) \cdot 0 = \pi_t \mathbf{J}_{\text{R}}(\phi_t).$$

Inverting the Cramér–Rao inequality gives $\text{CRB}_{\text{U}}(\phi_t) = \pi_t^{-1} \text{CRB}_{\text{R}}(\phi_t)$ per unit, and this ratio is preserved after aggregating over N units. Under MLE regularity, both $\hat{p}_{\text{U}}$ and $\hat{p}_{\text{R}}$ are asymptotically efficient, so their $N \cdot \text{MSE}$ limits equal their respective Cramér–Rao bounds; the ratio of these limits is (4). $\square$

Proof of part (ii). The CATE estimator $\hat{\tau}(x) = \hat{\mu}_1(x) - \hat{\mu}_0(x)$ is a linear functional of the two mean estimators; under MLE asymptotic normality the CR bound on $\text{Var}[\hat{\tau}]$ equals the sum of the per-arm CR bounds, since the two mean estimators are constructed from disjoint units and are asymptotically independent.

Write $v_t := \text{CRB}_R(\hat{\mu}_t)$ for the per-arm CRB on the arm- t mean under the joint objective. Part (i) with the mean as a smooth functional of the marginal density gives $\text{CRB}_U(\hat{\mu}_t) = \pi_t^{-1}v_t$, so

$$\frac{\text{Var}[\hat{\tau}^U]}{\text{Var}[\hat{\tau}^R]} \longrightarrow \frac{\pi_1^{-1}v_1 + \pi_0^{-1}v_0}{v_1 + v_0}.$$

Under the standing prior over SCMs the arms are exchangeable, so $v_0 = v_1 =: v$; substituting,

$$\frac{\text{Var}[\hat{\tau}^U]}{\text{Var}[\hat{\tau}^R]} \longrightarrow \frac{\pi_1^{-1} + \pi_0^{-1}}{2} = \frac{1}{2\pi(1-\pi)},$$

where the last equality uses $\pi_1^{-1} + \pi_0^{-1} = \pi^{-1} + (1-\pi)^{-1} = 1/(\pi(1-\pi))$. This is (5), and the special case $\pi = 1/2$ recovers the factor of 2 often quoted informally. $\square$

Remark 4. Equation (4) is an asymptotic-variance statement, not a finite-sample inequality. Under MLE regularity both $\mathbb{E}[(\hat{p}_U - p^*)^2]$ and $\mathbb{E}[(\hat{p}_R - p^*)^2]$ shrink as $\mathcal{O}(1/N)$; multiplying by N produces finite limiting constants C_U, C_R whose ratio equals π_t^{-1} .

B.3 Proof of Corollary 3

Proof. $\sqrt{\text{PEHE}}(x)$ is the square root of a monotone function of $\text{Var}[\hat{\tau}(x)]$. Under Theorem 2(ii) the variance ratio $\text{Var}[\hat{\tau}^R]/\text{Var}[\hat{\tau}^U] = 2\pi(1-\pi)$ holds pointwise in the SCM: the calculation depends only on the Fisher-information doubling from Lemma 1, which is a per-SCM statement. Consequently the ratio applies inside any second moment across SCMs, and in particular to the across-SCM variance of $\sqrt{\text{PEHE}}$:

$$\lim_{N \rightarrow \infty} \frac{\text{Std}_{\text{SCMs}}(\sqrt{\text{PEHE}}^R)}{\text{Std}_{\text{SCMs}}(\sqrt{\text{PEHE}}^U)} = \sqrt{2\pi(1-\pi)}.$$

$\square$

Remark 5 (Distribution-free scope). *The proofs invoke only Fisher information and MLE regularity; no Gaussianity, exponential-family, or homoscedasticity assumption is needed. Any joint law $\mathbb{P}^*$ with finite Fisher information for the parameters of the arm marginals satisfies both (4) and (5). The exchangeability step $v_0 = v_1$ in the proof of part (ii) can be dropped at the cost of stating the ratio in the unsimplified form $(\pi_1^{-1}v_1 + \pi_0^{-1}v_0)/(v_1 + v_0)$.*

C MALC

Section 3.2.1 recovers the per-unit CATE density by marginalising a continuous joint density $\hat{p}(y_0, y_1)$ fitted to the joint-head grid $\{p_{j_0, j_1}\}$ of (11). In this part we specify the details behind the estimator used for $\hat{p}$: a K -component mixture of mean-adjusted log-concave (MALC) densities, following the base construction of Danisman [2026]. The mixture is needed because posteriors over compatible SCMs can be multimodal, and a single log-concave density is unimodal by construction; the shape constraint is retained within each component while up to K modes can be represented across components.

Base class. Each mixture component is a single mean-adjusted log-concave (MALC) density $\hat{f} \in \mathcal{F}_{\text{LC}}$, where $\mathcal{F}_{\text{LC}}$ is the class of densities on $\mathbb{R}^2$ with concave log. We take the constructor, sampling proposal, and convergence guarantees of a single MALC fit as given by [Danisman \[2026\]](#).

Mixture family. For a fixed order $K \geq 1$, the mixture family is

$$\mathcal{F}_{\text{MALC}}^{(K)} = \left\{ f = \sum_{k=1}^K \pi_k f_k : \pi \in \Delta^{K-1}, f_k \in \mathcal{F}_{\text{LC}} \ \forall k \right\}. \quad (21)$$

Membership in $\mathcal{F}_{\text{MALC}}^{(K)}$ permits up to K modes while retaining the shape constraint componentwise.

Mixture EM on binned data. For $K > 1$, the components are combined by expectation–maximisation directly on the grid. Let $\gamma_{j_0 j_1, k}^{(t)}$ denote the responsibility of component k for bin (j_0, j_1) at iteration t , and $F_k^{(t)}(j_0, j_1) := \int_{B_{j_0, j_1}} f_k^{(t)}$ the bin-integrated component density. The updates are

$$\gamma_{j_0 j_1, k}^{(t+1)} = \frac{\pi_k^{(t)} F_k^{(t)}(j_0, j_1)}{\sum_l \pi_l^{(t)} F_l^{(t)}(j_0, j_1)}, \quad (22)$$

$$\pi_k^{(t+1)} = \sum_{j_0, j_1} p_{j_0, j_1} \gamma_{j_0 j_1, k}^{(t+1)}, \quad (23)$$

$$f_k^{(t+1)} = \text{MALC}\left(p_{j_0, j_1} \gamma_{j_0 j_1, k}^{(t+1)}\right), \quad (24)$$

Initialisation is a Voronoi assignment around the modal peaks of p , and the iterations stop after five consecutive non-improving log-likelihoods.

Order selection. The number of components is chosen by BIC on an effective sample size n_{eff} derived from the grid probabilities themselves:

$$n_{\text{eff}} := \left(\sum_{j_0, j_1} p_{j_0, j_1}^2 \right)^{-1}, \quad \text{BIC}(K) = -2 n_{\text{eff}} \sum_{j_0, j_1} p_{j_0, j_1} \log \hat{f}^{(K)}(j_0, j_1) + (3K - 1) \log n_{\text{eff}}, \quad (25)$$

where $\hat{f}^{(K)} \in \mathcal{F}_{\text{MALC}}^{(K)}$ is the mixture obtained by iterating (22)–(24) at order K . We select $K^* = \arg \min_{K \in \{1, \dots, K_{\text{max}}\}} \text{BIC}(K)$ with $K_{\text{max}} = 5$. The parameter count $3K - 1$ tallies the $K - 1$ mixture weights plus a two-dimensional mean per component; the log-concave shape itself does not contribute a finite parameter count. The synthetic-sample size $B = 500$ is used inside each component refit.

Each per-order fit is $\mathcal{O}(T \cdot K_g^2 + T \cdot K \cdot B \log B)$ where T is the EM iteration count (typically ≤ 20 in our experiments); the outer sweep over orders adds a factor K_{max} . Because the fit uses the binned probabilities directly, no resampling of the original context is required and the whole procedure runs on a single CPU per query.

Output and downstream use. The MALC estimator is a smooth density

$$\hat{p}_{\text{MALC}}(y_0, y_1) = \sum_{k=1}^{K^*} \hat{\pi}_k^* \hat{f}_k^*(y_0, y_1) \quad (26)$$

on the convex hull of the synthetic points that generated the components, evaluable at any query $(y_0, y_1) \in \mathbb{R}^2$.

D Optimal-transport aggregation

Section 3.2.2 defines the ATE density f_{ATE} as the uniformly-weighted 2-Wasserstein barycenter of the per-query CATE densities $\{f_\tau^{(n)}\}_{n=1}^N$ and states its closed-form quantile representation in (7). In this segment we report the derivation and the grid-level algorithm used for ATE.

Notation. Let F_n and $Q_n := F_n^{-1}$ be the CDF and quantile function of the per-query CATE density $f_\tau^{(n)}$, and let $\alpha \in \Delta^{N-1}$ be nonnegative weights summing to 1. On $\mathbb{R}$ the weighted 2-Wasserstein barycenter of $\{f_\tau^{(n)}\}_{n=1}^N$ admits the closed-form quantile representation

$$Q^*(t) = \sum_{n=1}^N \alpha_n Q_n(t), \quad t \in (0, 1), \quad (27)$$

of which (7) is the uniform-weight case $\alpha_n = 1/N$.

Grid-level implementation. Because Q^* is defined explicitly by (27), no iterative optimisation is needed. Given the per-query densities sampled on a common grid $x_1 < \dots < x_M$ with spacing Δx , the barycenter density is produced in three quantile passes and one density-recovery pass:

1. Discretise the probability axis by $\tau_j := (j - \frac{1}{2})/M_\tau$ for $j = 1, \dots, M_\tau$ (we use $M_\tau = 4001$).
2. For each query $n \in \{1, \dots, N\}$, compute the CDF by trapezoidal integration, renormalise, and invert to obtain the quantile function at every τ_j :

$$F_n(x_k) \leftarrow \sum_{\ell \leq k} \frac{1}{2} (\hat{p}_n(x_{\ell-1}) + \hat{p}_n(x_\ell)) \Delta x, \quad F_n \leftarrow F_n / F_n(x_M), \quad Q_n(\tau_j) \leftarrow F_n^{-1}(\tau_j).$$

3. Average the quantile functions pointwise per (27): $Q^*(\tau_j) = \sum_n \alpha_n Q_n(\tau_j)$.
4. Recover the barycenter density on the original grid: invert Q^* back to a CDF F^* , take a central-difference derivative, and renormalise to unit integral,

$$\hat{f}_{\text{ATE}}(x_k) \leftarrow \frac{F^*(x_{k+1}) - F^*(x_{k-1})}{2 \Delta x}, \quad \hat{f}_{\text{ATE}} \leftarrow \hat{f}_{\text{ATE}} / \left(\Delta x \sum_k \hat{f}_{\text{ATE}}(x_k) \right).$$

Every step is $\mathcal{O}(N \cdot M)$ vector arithmetic; producing $\hat{f}_{\text{ATE}}$ is therefore much cheaper than the per-query MALC fits it aggregates.

E Theory Validation something something

References

- A. W. van der Vaart. *Asymptotic Statistics*. Cambridge University Press, 1998.
- W. F. Sheppard. On the application of the theory of error to cases of normal distribution and normal correlation. *Philosophical Transactions of the Royal Society A*, 192:101–167, 1899.
- F. Danisman. Mean-adjusted log-concave density estimation, 2026. Working paper.